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Paleomagnetism of certain constituents of the Bay St. George sub-basin, western Newfoundland

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Paleomagnetic studies have been made of certain constituents of the Bay St. George sub-basin. Specifically, results are reported from the Spout Falls Formation (Tournaisian), the Jeffreys Village Member of the Robinsons River Formation (Visean), and the Searston Formation (Namurian–Westphalian). The following magnetizations have been isolated: Spout Falls A (Tournaisian) with $D = 343.5°$, $I = -22.7°$, $k = 61.2$, $\alpha_{95} = 7.1°$ and the corresponding pole at 28.6°N, 139.5°E (4.5°, 8.5°); Spout Falls B (Kiaman) with $D = 166.7°$, $I = 12.2°$, $k = 51.7$, $\alpha_{95} = 10.7°$ and the corresponding pole at 34.5°S, 42.7°W (5.5°, 10.9°); Jeffreys Village A (Visean) with $D = 351.2°$, $I = -27.3°$, $k = 54.0$, $\alpha_{95} = 7.6°$ and the corresponding pole at 26.5°N, 130.7°E (4.5°, 8.3°); Searston A (Namurian) with $D = 161.7°$, $I = 11.7°$, $k = 107$, $\alpha_{95} = 7.4°$ and the corresponding pole at 33.9°S, 37.2°W (3.8°, 7.5°); and Searston C with $D = 111.6°$, $I = -13.8°$, $k = 28.8$, $\alpha_{95} = 14.5°$ and the corresponding pole at 19.6°S, 19.0°E (7.6°, 14.8°). After comparison with paleopoles of similar ages derived from eastern and western Newfoundland rocks, from constituents of the east coast basin and for interior North America, it is concluded that: (1) it is unlikely that any large scale relative motion took place since the Early Carboniferous between eastern and western Newfoundland; (2) it is unlikely that any north–south relative motion took place between the east coast basin and the Bay St. George sub-basin; and (3) the Bay St. George sub-basin results do not support the earlier proposed displaced terrane hypothesis of the northern Appalachians in as much as the motions *during* the Carboniferous are not supported. There is evidence of the northward motion of the Appalachians and North America *as a whole* during the Carboniferous. The magnetostratigraphic horizon marker in the Carboniferous separating a dominant normal and reversed magnetization on the older side and an entirely reversed (Kiaman) magnetization on the younger side may be placed in the Bay St. George sub-basin at the base of the Searston Formation.

1. Objectives of the study

Paleomagnetism of certain constituents of the Bay St. George sub-basin of western Newfoundland is attempted. The objective is to establish Carboniferous paleopoles relative to this part of western Newfoundland. Paleomagnetic data from the Carboniferous of western Newfoundland have important tectonic and magnetostratigraphic implications: (1) if the data from eastern and western Newfoundland agree, such agreement precludes any relative motion between the two regions since the Carboniferous. At present, the available Carboniferous data from eastern Newfoundland are of the Terrenceville Formation by Kent (1982), and of the St. Lawrence pluton by Irving and Strong (1984a) while from the west, the Visean Deer Lake Group was studied (Irving and Strong, 1984b). A preliminary result from the Barachois "Fm." (Opdyke, 1981) is available in abstract form; (2) the data will have a bearing on the displaced terrane hypothesis of Kent and Opdyke. Devonian and Carboniferous paleopoles of the Maritime–New England region of Canada and U.S. were claimed to differ from those of cratonic North America (Kent and Opdyke, 1978, 1979,

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1980; Irving, 1979). This led to suggestions (Kent and Opdyke, 1978, 1979, 1980) of a major left-lateral displacement by about 1500 km of an exotic terrane called Acadia relative to North America during the Devonian and Carboniferous times. However, Roy (1982) called attention to the possibility of unresolved overprinting of cratonic data for the Devonian and the Carboniferous and hence to the fragile basis of the Kent–Opdyke hypothesis. The displaced terrane hypothesis was refuted by Irving and Strong (1984b,c) who showed analytically that the cratonic magnetizations were indeed overprints similar to the overprints found in their Deer Lake data. They also showed that the same Early Carboniferous paleolatitudes were obtained both for eastern and western Newfoundland disproving the Kent–Opdyke hypothesis. If the primary magnetizations of the constituents of the Bay St. George sub-basin agree with what were described in the literature as Mississippian magnetizations of the cratonic North American rocks, it implies that the data for cratonic North America are also true representatives of the Mississippian magnetic field and the contention that the Kiaman superpositions have not been separated out for the cratonic data (Irving and Strong, 1984c) may be discounted; (3) perfect agreement between western Newfoundland and cratonic North American poles also precludes the suggestion of the rotation of western Newfoundland relative to mainland North America; and (4) Roy and Morris (1983) suggested that a magnetostratigraphic horizon may be marked in the east coast sedimentary basin of North America, the marker separating predominantly normal and reversed magnetizations on the older side and an entirely reversed group on the younger side. They place this horizon marker at Westphalian A. The Formations of the Bay St. George sub-basin encompass this age and it would be of interest to pinpoint this horizon marker in the sub-basin for correlative purposes.

Early paleomagnetic studies of the constituents of the Bay St. George sub-basin include a study of nine samples by Nairn et al. (1959), of 32 samples by Black (1964) and of at least 50 samples of the Barachois Formation by Opdyke (1981). Of these, Nairn et al. (1959) reported only NRM data without any demagnetizations, and Black conducted only AF demagnetizations upto 30 mT. It was not clear from Black's paper either from which formation of the sub-basin his samples were collected, or whether any structural corrections were applied for the data. Black's work was done prior to the recognition of the possible multi-component magnetizations and prior to the advent of the many analytical and experimental techniques now available for resolving magnetizations in a sample. Further, AF demagnetization in a 30 mT field is not adequate to isolate primary magnetization in a sedimentary rock unit with hematite as the probable remanence carrier. In spite of such drawbacks, the results by Black are invariably cited in any discussion of the Appalachian paleomagnetism and tectonics. Opdyke (1981) reported results from at least 50 samples of 300 m thickness of the Barachois (Fm.?) Group of the sub-basin. The result is in abstract form only and many details are not available to make meaningful conclusions of the objectives listed above. Thus, it was necessary to make a thorough paleomagnetic study of the constituents of the sub-basin.

2. Geology

The Carboniferous rocks of the Bay St. George sub-basin have been studied geologically for a long time (Hays and Johnson, 1938; Bell, 1948; Riley, 1962; Baird and Cote, 1964). Recent studies included a comparison of this sub-basin with the Maritime sedimentary basin by Belt (1969) and a very recent thorough re-examination of all geologic and stratigraphic aspects of the sub-basin by Knight (1983). Although the nomenclature has not been changed greatly since the beginning of the studies, the names of the stratigraphic units as suggested by Knight (1983) are followed in this paper.

The northeasterly trending Bay St. George sub-basin [Fig. 1] is considered an extension of the larger Maritimes Basin and is said by Knight to contain ~ 10 km thickness of sediments ranging in age from Late Devonian (?) to Late Carboniferous. The strata are divided into three Groups: the Anguille Group (Late Devonian (?) to Tournaisian), Codroy Group (Visean) and the Barachois

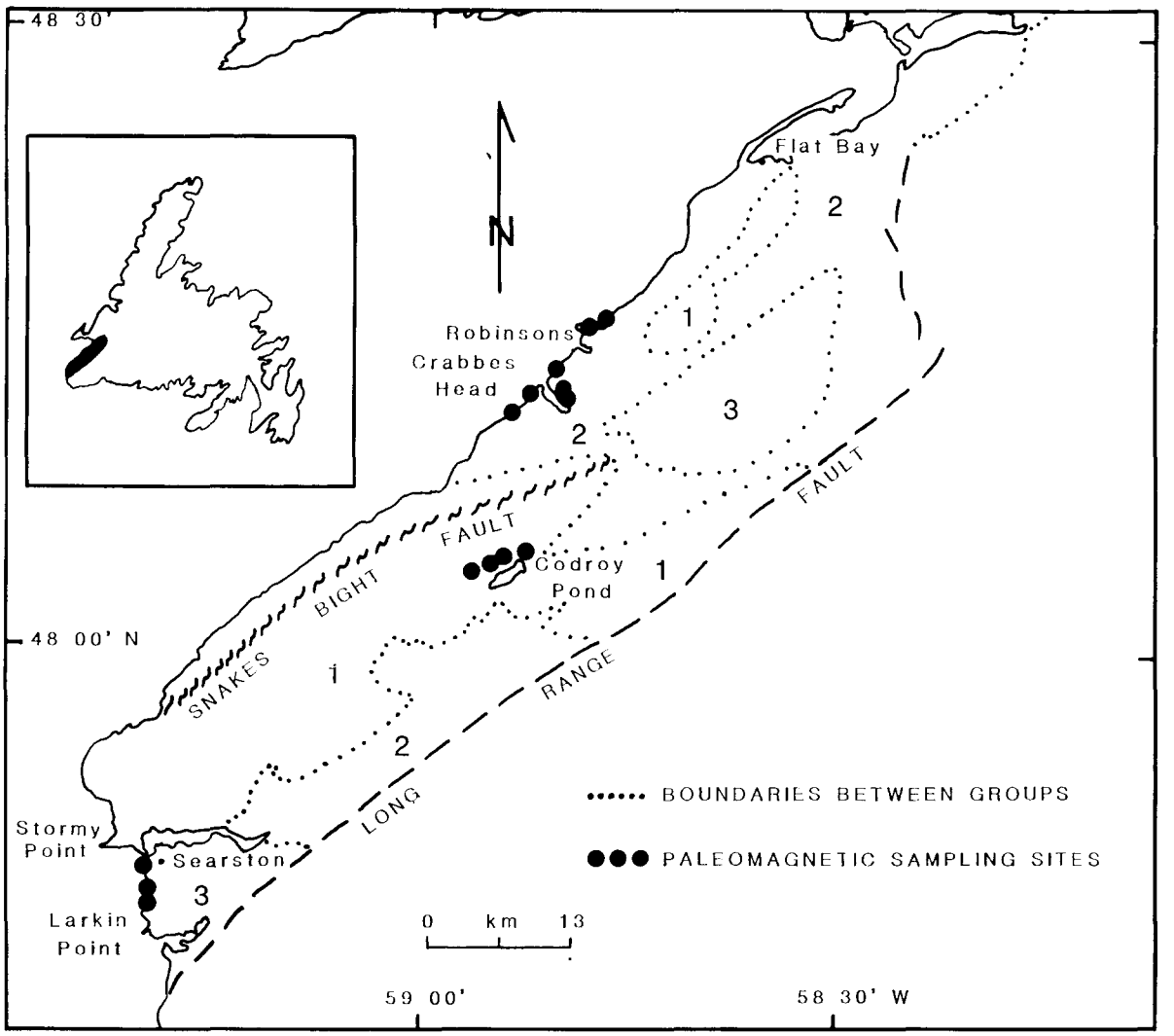


Fig. 1. Geological sketch map of the Bay St. George sub-basin with the map of Newfoundland as inset (modified from Knight, 1983). 1: Anguille Group; 2: Codroy Group; (3) Barachois Group.

Group (Namurian–Westphalian). They are considered broadly equivalent both in lithology and time to Horton, Windsor and Canso/Riversdale Groups, respectively, of the Maritimes Basin. The Maritimes Basin is a successor basin formed after the completion of the Acadian deformation of the Appalachian fold belt, uplift of the fold belt probably starting during or soon after the Early Devonian. Rocks of the Maritimes Basin were deformed in the Late Carboniferous. Most of the constituents of the Bay St. George sub-basin are non-marine terrigeneous clastic sediments with main marine incursions in the Codroy Group, corresponding to Late Visean marine transgressions which occurred throughout North Africa and Europe (Schenk, 1971). The basement rocks of the sub-basin comprise to the northwest, the Humber Zone of the Newfoundland Appalachians and to the southeast, the rocks of the Dunnage and Gander Zones. Figure 2 shows the stratigraphic

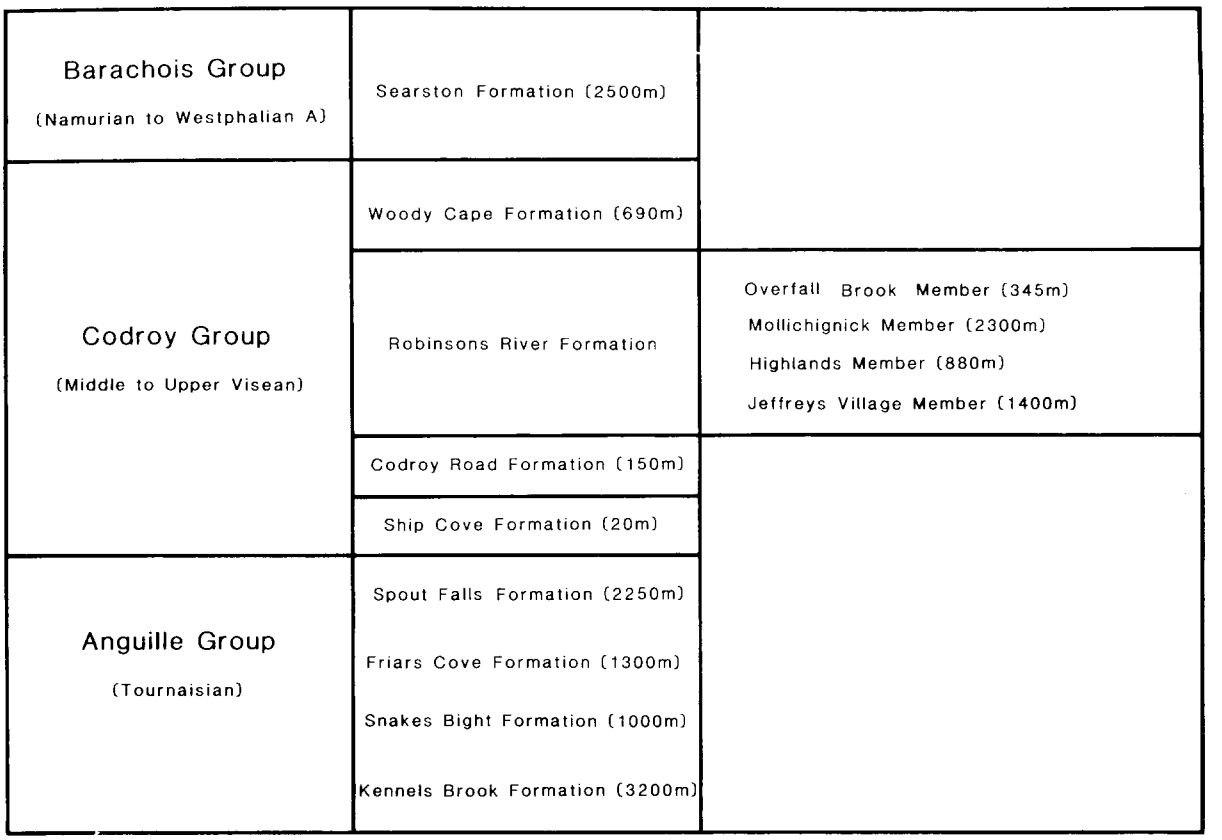


Fig. 2. Stratigraphic column in the Bay St. George sub-basin with the thickness of each Formation indicated in meters.

subdivisions in the sub-basin, and their estimated thicknesses.

Of the formations listed in Fig. 2, results from the Spout Falls, Robinsons River and the Searston are presented in this paper.

2.1. Spout Falls Formation

The Spout Falls Formation comprises red and gray sandstones with minor siltstones and conglomerates. These rocks occur in the northern half of the Anguille mountains, Bald mountain and continue northeast upto Barachois Brook. While the exposures are plentiful in most streams northwest of the Snakes Bight Fault, most of the paleomagnetic sampling was done in a road cut and surrounding areas (microwave tower) north of the Codroy pond. The sampling areas are dominated by red sandstones exhibiting a sheet-like geometry. Similar redbed sequences as the Spout Falls Formation are described near the top of the Horton Group in the Maritimes Basin where they underlie the base of the Windsor Group. Possible correlatives of the Spout Falls Formation in the Maritimes Basin are the Cheverie Formation of Nova Scotia (Bell, 1960) and the Ainslie Member of the Ainslie–Strathlorne Formation of Cape Breton (Kelley, 1967).

2.2. Robinsons River Formation

The Robinsons River Formation of the Codroy Group underlies both the St. Georges Bay and the Codroy lowlands. Knight (1983) subdivided this Formation into four Members [Fig. 2] based upon lithological characteristics and biostratigraphic in-

formation. The Jeffreys Village Member, the lowermost of the four, conformably overlies the Codroy Road Formation and consists of red and gray shales, mudstones, siltstones, sandstones and conglomerates interbedded with carbonates and evaporites. Sampling in this member is confined to red sandstones and siltstones, the sandstones exhibiting cross-bedding and planar stratification. This member was sampled between Crabbes Head and Robinsons along the coast. The Codroy Group is Late Mississippian in age and is correlated with the Windsor Group of Nova Scotia which is considered to be Middle to Late Visean age.

2.3. Searston Formation

The Searston Formation of the Barachois Group is the youngest rock unit in the sub-basin and consists of a thick succession of gray sandstones intercalated with red siltstones. This formation was sampled along the coast between Kennedy Point and Larkin Point. Most of the beds are organized into fining-upward sequences up to 60 m thick. Thick green-gray to red sandstones form at least 50% of the Formation, and have characteristics that conform to those generally attributed to those of river channels. Red siltstones, which comprise about 35% of the formation, are interbedded with thin red sandstones and intraformational conglomerates. Knight (1983) provided arguments for this formation to be near the Visean–Namurian boundary. Paleocurrent and petrographic studies of the sandstones indicate that the source of the sediments are the mountains southeast of the Long Range fault.

3. Paleomagnetism

A total of 88 independently oriented samples were collected from 21 sites covering the Spout Falls (eight sites), the Robinsons River (eight sites) and the Searston (five sites) Formations. Of the Robinsons River Formation, Jeffreys Village Member was the only unit sampled. The sampling localities are shown in Fig. 1. Oriented samples were taken either from red sandstones or red siltstones. Strikes and dips were carefully measured at each site and are presented in Table I. At least two specimens, each 2.5 cm in length, were cut from each sample and their remanent magnetization was measured using a Schonstedt SSM-1 spinner magnetometer. The remanent intensities are fairly homogeneous and are of the order of 10^{-3} A m^{-1}. Thermal demagnetizations were conducted of each specimen at a minimum of 13 steps up to 680°C

TABLE I

Attitudes of beds at paleomagnetic sampling sites

Site	Strike and dip [1]
Spout Falls Formation	
A D110–D114	355°, 48°E
B D115–D118	2°, 51°E
C D119–D124	36°, 74°E
D D125–D129	23°, 73°E
E D130–D133	24°, 55°E
F D134–D139	15°, 61°E
G D140–D143	32°, 62°E
H D144–D147	26°, 64°E
Robinsons River Formation (Jeffreys Village Member)	
I D25–D30	200°, 61°NW
J D31–D34	205°, 58°NW
K D35–D37	225°, 56°NW
L D38–D42	197°, 57°NW
M D43–D47	216°, 66°NW
N D48–D50	208°, 58°NW
O D51–D55	206°, 59°NW
P D56–D58	201°, 62°NW
Searston Formation	
Q D66–D70	73°, 66°S
	54°, 54°S
	58°, 54°S
	54°, 55°S
	49°, 45°S
R D71–D74	60°, 57°S
	60°, 52°S
	56°, 59°S
	60°, 64°S
S D73–D77	60°, 60°S
	63°, 60°S
	63°, 59°S
T D78–D81	60°, 46°S
	60°, 46°S
	60°, 75°S
	60°, 67°S
U D82–D84	54°, 0°
	60°, 24°S
	60°, 25°S

[1] Dip direction is, by convention, 90° to the right of strike.

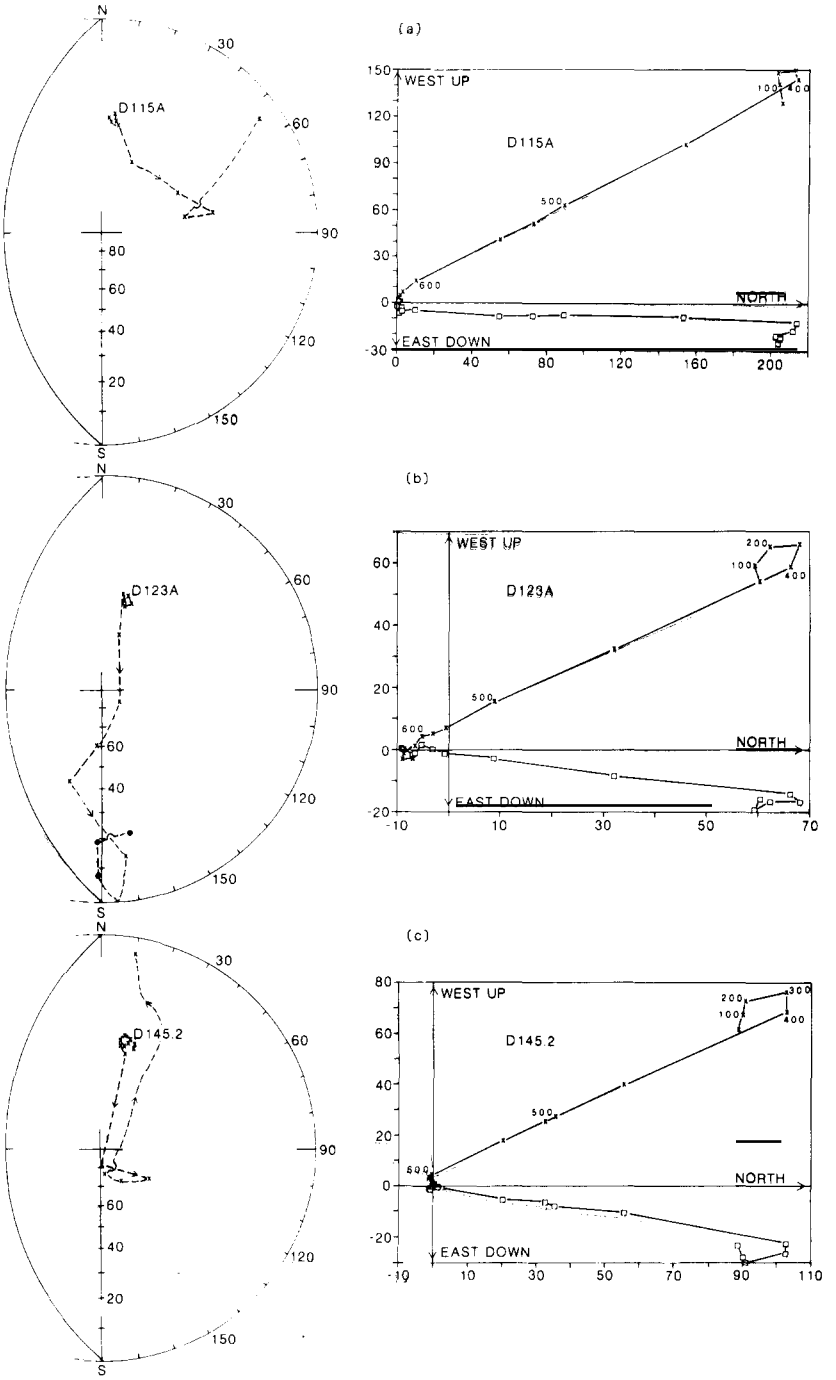


Fig. 3. Response to thermal demagnetization of specimens from the Spout Falls Formation presented in the form of stereonet plots (left) and Zijderveld diagrams (right). Solid symbols (crosses) on the stereonet plot indicate positive (negative) inclinations. For Zijderveld diagrams, open squares are projections of the vector end-points on the horizontal plane and crosses are projections on the vertical plane. Numbers next to crosses are the levels in °C of the thermal demagnetization step. Magnetization units on the axes are in 10^{-3} A m^{-1}. All the directional plots are with respect to the present horizontal.

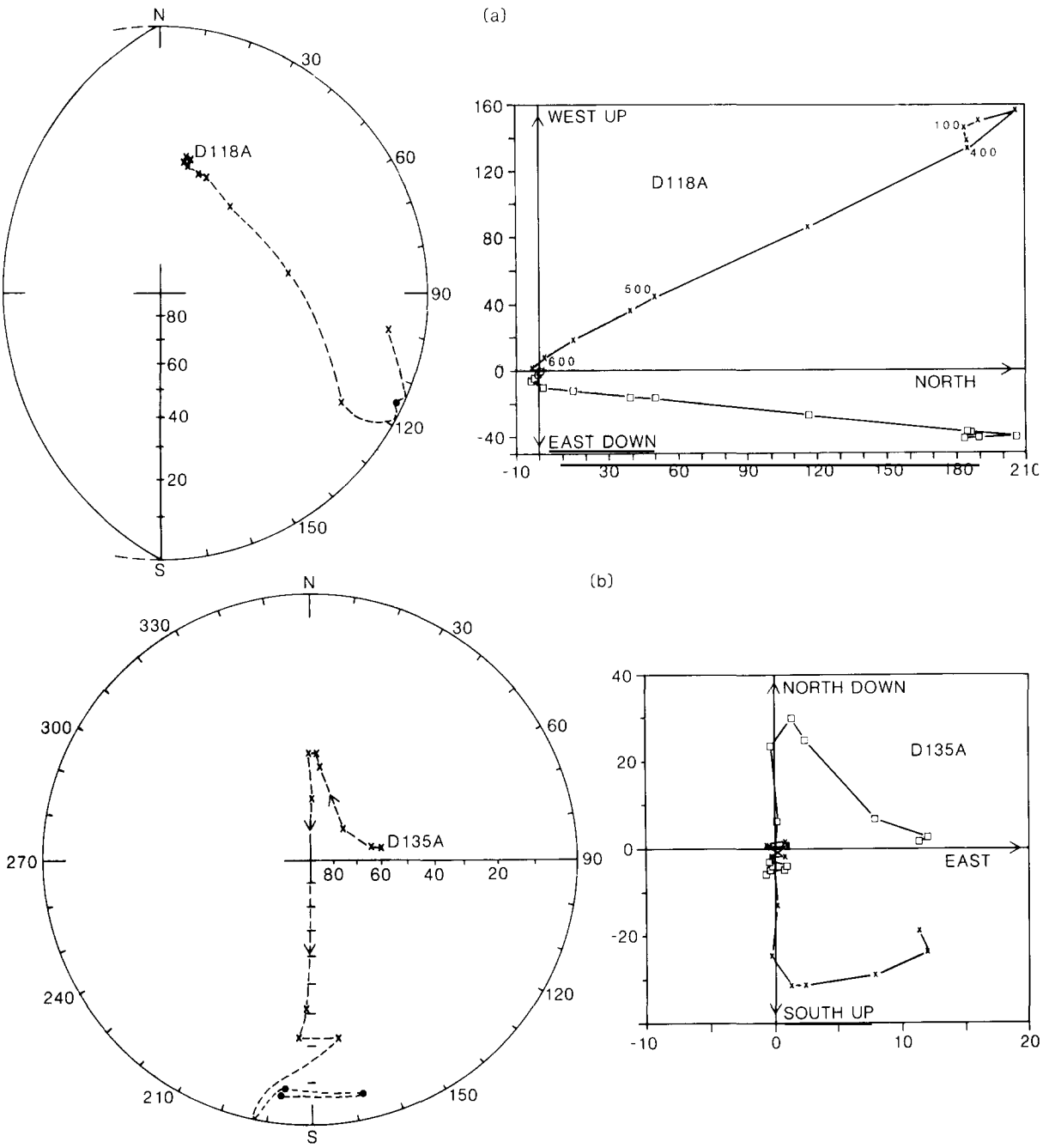


Fig. 4. Responses to thermal demagnetization of specimens from the Spout Falls Formation presented in the form of stereonet plots (left) and Zijderveld diagrams (right). The symbols have the same meaning as in Fig. 3. Magnetization units on axes are in 10^{-3} A m^{-1}. All directional plots are with respect to the present horizontal.

using a Schonstedt thermal demagnetizer in ambient fields of not more than 5 nT. The thirteen steps used were 100°, 200°, 300°, 400°, 450°, 500°, 530°, 560°, 590°, 600°, 620°, 640°, 660°, and 680°C. These steps are considered close enough to separate various components of magnetization. Zijderveld diagrams (Zijderveld, 1975; Dunlop, 1979) are exclusively used for identifying the various components. Most samples showed linear demagnetization segments over many demagnetization steps making it possible to identify the characteristic magnetization without any ambiguity.

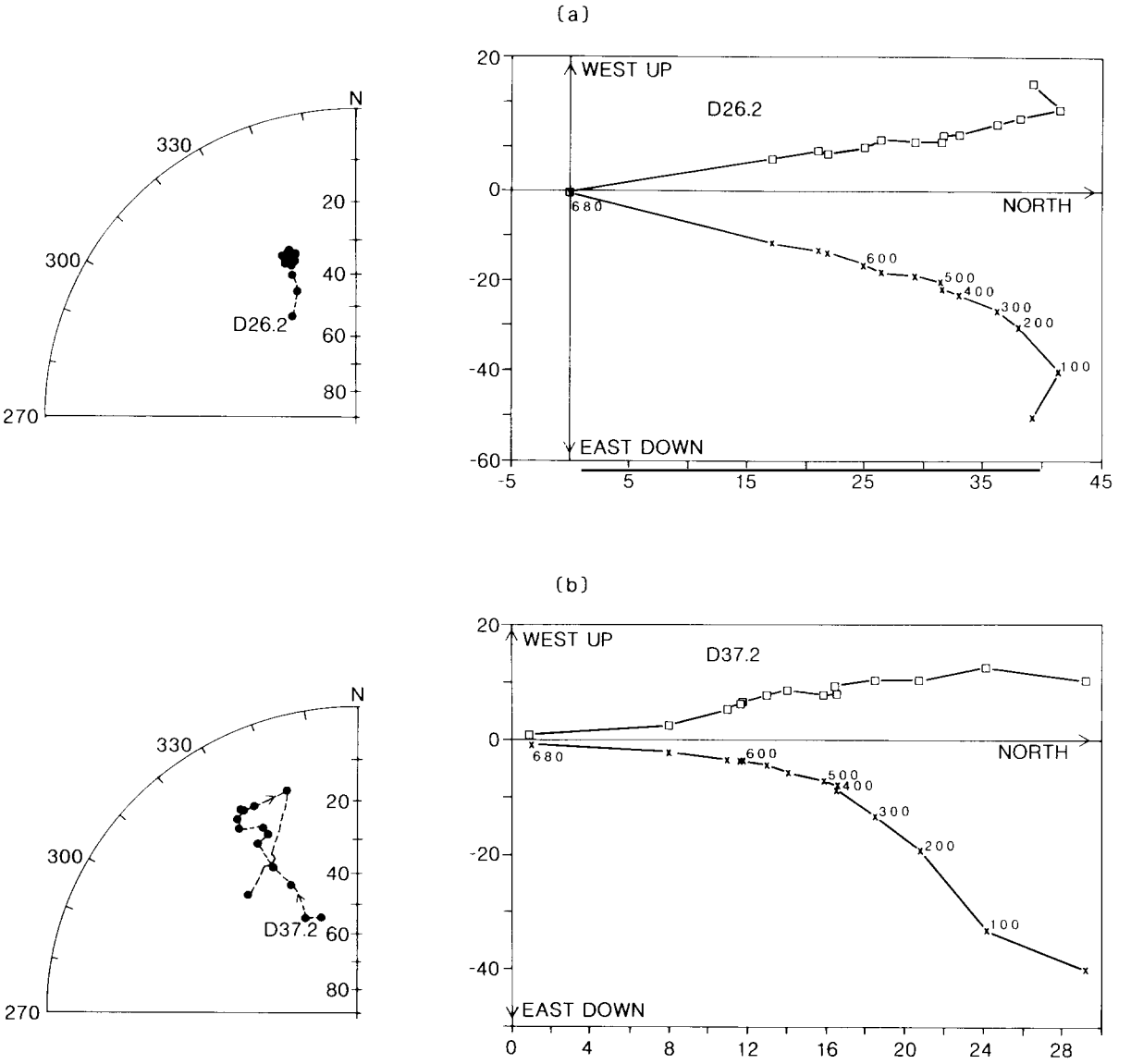
Zijderveld diagrams for the Spout Falls Formation are shown in Figs. 3 and 4. It is to be noted that, unlike the standard practice, the scales on the abscissa and the ordinates of the Zijderveld plots presented in this paper are not the same. This procedure was adopted as it facilitates identification of the linear segments more easily. In the first two demagnetization steps, a soft magnetization, probably the effect of the present Earth's field, is removed in all the specimens. In some specimens, of which D118A and D135A are typical (Fig. 4), magnetization directed toward southeast and shallow positive inclination is also present. Such specimens constitute 60% of the total Spout Falls collection. The magnetizations were obtained by calculating the slope of the line. For the characteristic magnetizations, it was noted that there are at least five and in many cases more data-points on the line. This method of obtaining the magnetization is equivalent to determining the vector differences in that demagnetization range. The dominant magnetization (which is probably the characteristic magnetization) of all the Spout Falls specimens is northerly directed and is upward. This characteristic magnetization seems to converge to the origin in most cases (Figs. 3 and 4). In cases where the vector end-points do not ap-

TABLE II

Isolated magnetizations for the Spout Falls Formation

Site	N	n	n'/n	Insitu				Tilt corrected			
				D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)
Magnetization A											
A	5	10	4/10	4.6	−28.0	43.3	14.1	338.7	−24.9	42.6	14.2
B	4	8	8/8	4.1	−40.4	26.7	18.1	329.5	−25.3	27.5	17.8
C	6	12	12/12	10.4	−33.3	85.5	7.3	339.4	−26.6	91.5	7.0
D	5	10	5/12	32.8	−30.7	12.5	22.6	355.5	−17.1	12.9	22.1
E	4	8	8/8	24.2	−41.5	58.7	12.1	347.3	−22.7	70.8	11.0
F	5	10	10/10	4.4	−45.0	48.4	11.1	332.3	−13.5	43.6	11.7
G	4	8	4/8	40.0	−42.5	11.7	28.1	356.0	−24.2	10.4	30.0
H	4	8	8/8	37.2	−42.5	101	9.2	348.9	−24.8	95.4	9.4
Mean	8 sites			19.5	−38.9	35.2	9.5	343.5	−22.7	61.26	7.1
Magnetization B											
A	5	10	10/10	162.1	10.0	36.8	12.8	158.4	−2.1	39.2	12.4
B	5	10	0/10								
C	6	12	8/12	172.6	27.2	24.4	19.0	171.8	−26.3	24.6	18.9
D	5	10	10/10	172.7	2.3	106	7.5	190.7	−28.6	108	7.4
E	5	10	0/10								
F	5	10	8/10	167.1	15.0	24.0	19.1	166.8	−14.4	22.5	19.8
G	4	8	6/8	159.4	6.2	16.4	31.5	173.7	−39.5	15.6	32.4
H	4	8	0/8								
Mean	5 sites			166.7	12.2	51.7	10.7	171.7	−22.6	20.3	17.4

N = number of samples; n = number of specimens used in demagnetization; n'/n = number of specimens showing the magnetization/-number of specimens used in demagnetization; D = declination; I = inclination; k = Fisher's precision parameter; α_{95} = radius of the 95% circle of confidence.



parently converge to the origin (e.g., D123A, Fig. 3), magnification of that part of the Zijderveld plot does not show any definite trend. The isolated magnetizations for the Spout Falls Formation with respect to the present horizontal and with respect to the bedding plane are listed in Table II.

Figure 5 shows the Zijderveld diagrams and demagnetization histories for typical specimens from the Jeffreys Village Member of the Robinsons River Formation. Effects of the present Earth's field were removed during the first two demagnetization steps (D52.1, D49.1, D26.2). A characteristic northwest magnetization with shallow inclinations (downward for some specimens as in examples D49.1, D52.1) can be identified. The vector end-points converge nicely to the origin. Stable end-points are noted in the stereonet plots. Save for the effect of the present Earth's field, the only magnetization present is the northwest magnetization with shallow inclinations. An exception

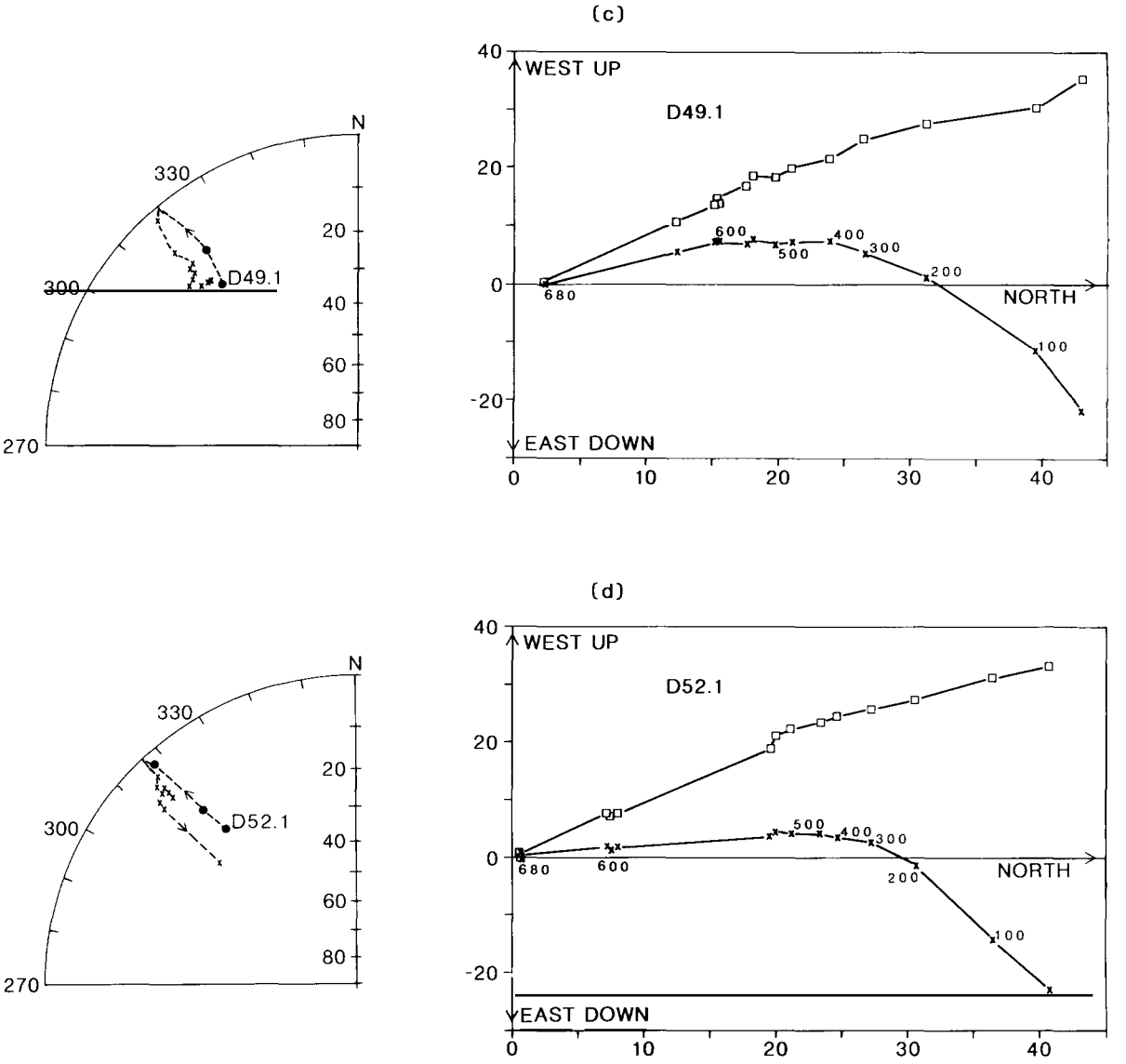


Fig. 5. Response to thermal demagnetization of four specimens from the Jeffreys Village Member presented in the form of stereonet plots (left) and Zijderveld diagrams (right) for each specimen. Symbols have the same meaning as in Fig. 3. Magnetization units on axes are in 10^{-3} A m^{-1}. All directional plots are with respect to the present horizontal.

to the above behavior was noticed in site P which is the bottommost bed exposed at Crabbls Head of the Jeffreys Village Member. The typical behavior of this set of specimens is shown in Fig. 6. First, in steps up to 300°C, a northerly, steep downward magnetization (probably the effect of the present Earth's field) was removed. From 300 to 660°C, a southerly magnetization with shallow positive inclination was removed. A blow-up of the vector end-point plot between 400 and 680°C (Fig. 6c) shows a few points straying off the straight line segment, but the trend can easily be recognized.

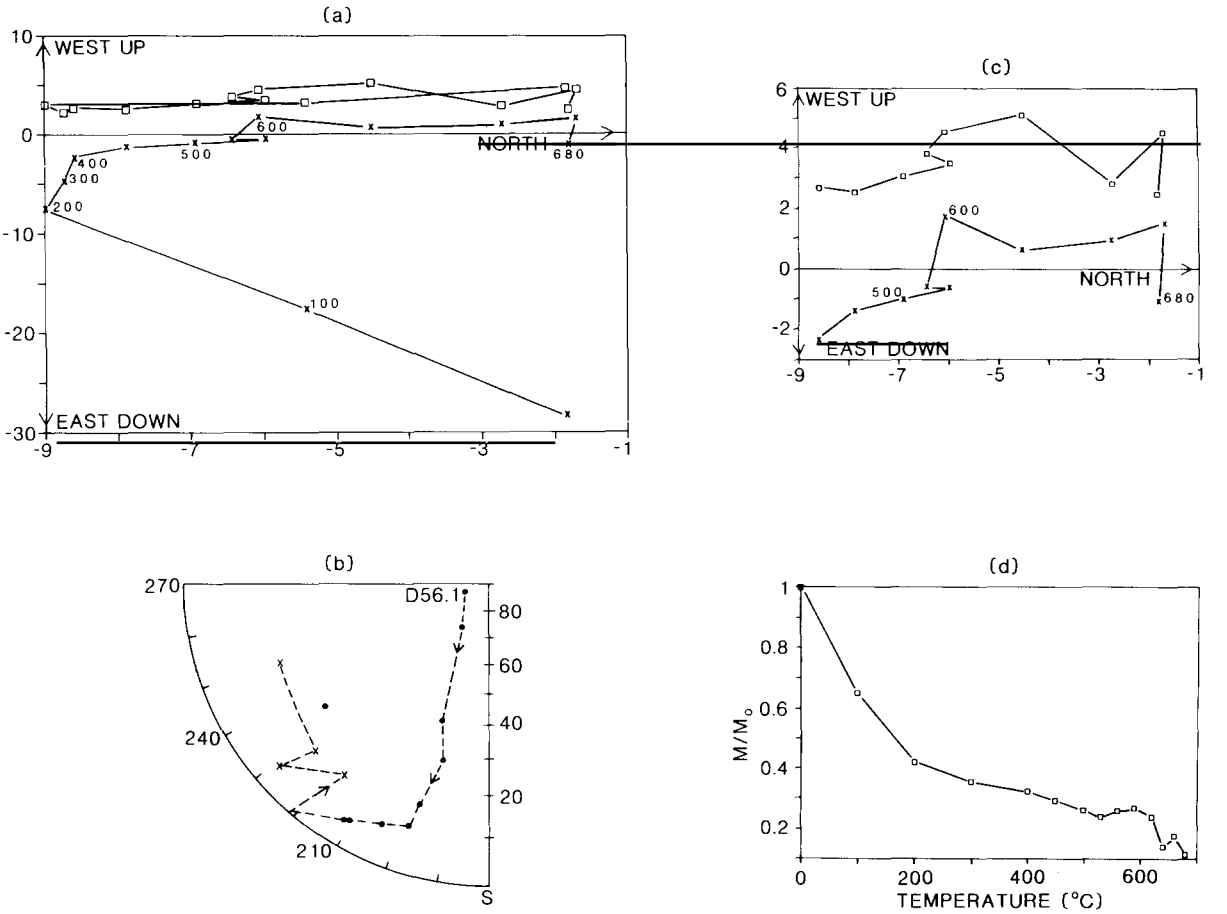


Fig. 6. Response of specimen D56.1 (Jeffreys Village Member) to thermal demagnetization. Symbols have same meaning as in Fig. 3. (a) Total Zijderveld diagram; (b) corresponding stereonet plot; (c) blown-up Zijderveld diagram in the range 400°C to 680°C; (d) normalized intensity during thermal demagnetization, versus temperature.

No other magnetization seems to be present. The corresponding stereo-net plot (Fig. 6b) and the fall of normalized intensity with temperature (Fig. 6d) are also shown. All three samples collected from this thin bed of red sandstone (total 10 specimens demagnetized from this site) behaved in a very similar fashion. Table III shows the magnetizations isolated for the Jeffreys Village Member.

Specimens of the Searston Formation showed a distinct behavior during demagnetization as shown by examples in Fig. 7. At low temperatures (up to 400°C), a southerly directed steep magnetization was removed. Beyond 400°C, a magnetization east–southeasterly directed with shallow (mostly negative) inclinations was recognized. The vector end-points converged to the origin indicating these to be the only magnetizations present in the specimens. Both magnetizations are present in all the specimens. Response of specimens to demagnetization is much more uniform for this formation, all

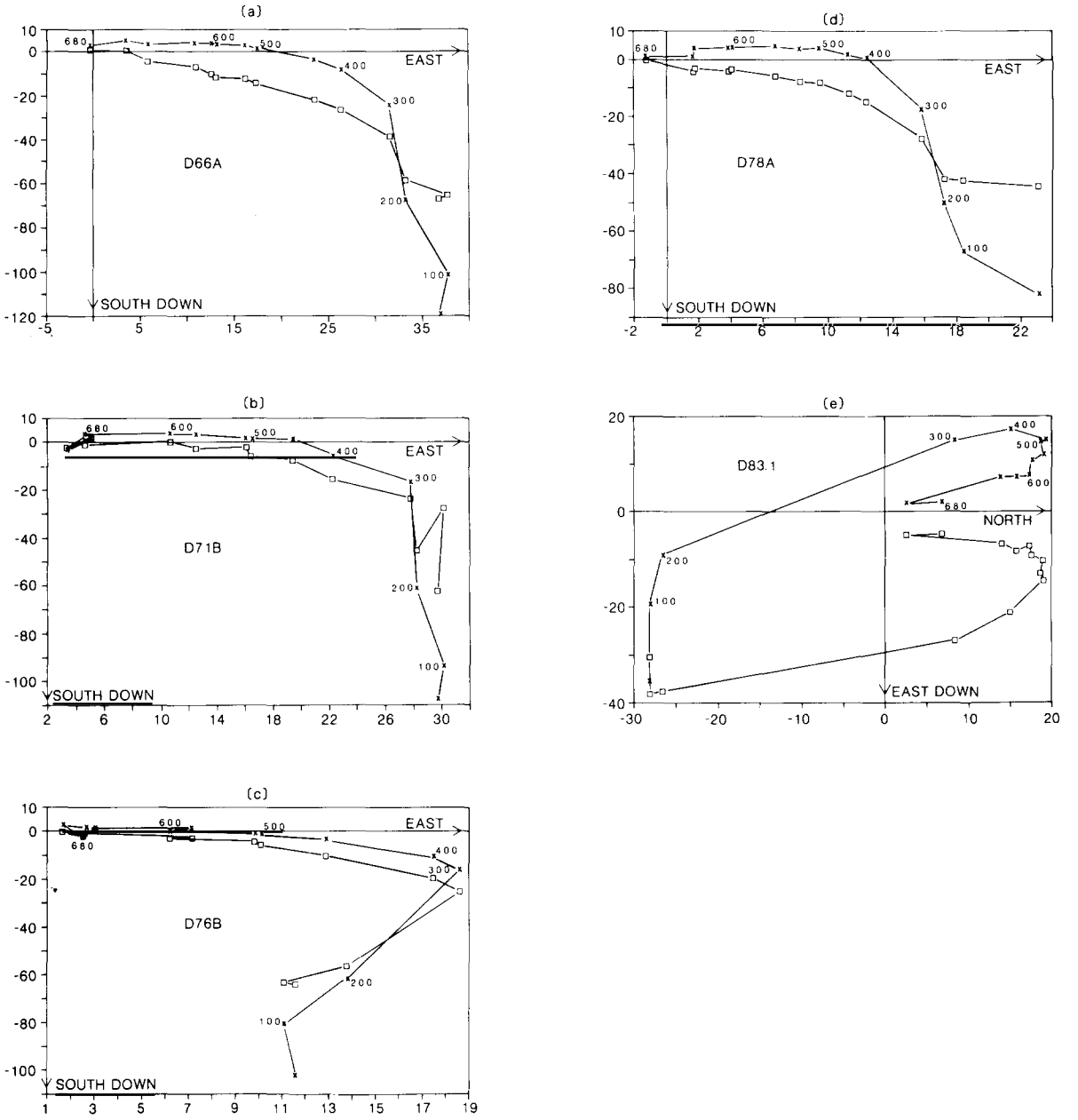


Fig. 7. Response to thermal demagnetization of five specimens from the Searston Formation presented in the form of Zijderveld diagrams. Magnetization units on axes are in 10^{-3} A m^{-1}. Symbols have the same meaning as in Fig. 3. All the plots are with respect to the present horizontal.

specimens invariably behaving in the same fashion. The isolated magnetizations for the Searston Formation, both in-situ and after tilt correction, are presented in Table IV.

TABLE III

Isolated magnetizations for the Jeffreys Village Member (Robinsons River Formation)

Site	N	n	n'/n	In situ				Tilt corrected			
				D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)
Magnetization A											
I	5	10	10/10	348.0	12.0	17.3	18.9	352.1	−20.1	17.4	18.9
J	3	6	6/6	348.4	9.5	12.9	35.8	355.1	−23.4	16.7	31.2
K	3	6	6/6	333.7	16.4	19.4	28.8	337.0	−36.7	19.8	28.5
L	5	10	10/10	357.1	−6.3	6.0	34.2	10.9	−20.0	6.0	34.2
M	5	10	10/10	349.8	23.3	14.1	21.1	351.0	−26.5	13.9	21.2
N	3	6	6/6	334.7	12.0	52.4	6.6	342.6	−33.1	51.6	6.8
O	5	10	10/10	336.9	2.1	23.9	11.2	343.1	−30.7	22.4	12.6
P	3	6	6/6	166.7	−8.6	33.6	21.6	174.8	25.1	25.8	24.0
Mean *	8 sites			344.4	11.1	48.1	8.1	351.2	−27.3	54.0	7.6

N = number of samples; n = number of specimens used in demagnetization; n'/n = number of specimens showing the magnetization/number of specimens used in demagnetization; D = declination; I = inclination; k = Fisher's precision parameter; α_{95} = radius of the 95% circle of confidence; * direction of site P is reversed in calculating the mean.

4. Analysis of results

It may be said with reasonable certainty that the magnetizations listed in Tables II–IV are the only magnetizations present in the samples (apart from soft present field effects). It can also be stated with confidence that these magnetizations are real and are not the result of the techniques of analysis used and can be seen without ambiguity in the Zijderveld diagrams. Now, it needs to be ascertained which of these magnetizations were acquired with the beds in their present attitude (post-tilting) and which were acquired with the beds horizontal (probably primary). For the Spout Falls Formation, what is termed magnetization A (Table II) is probably a pre-tilting magnetization. The slightly different structural corrections at various sites may be utilized so that the statistics of the calculated means may yield a modified fold test, viz. if the in situ precision parameter is lower than the tilt corrected precision parameter, then it may be stated with reasonable confidence that the magnetization was probably acquired before tilting took place. For the case of Spout Falls A, although improvement in Fisher's k with tilt correction (Table II) is not at a significant level as per McElhinny's (1964) calculations there is improvement in grouping with structural correction. As McFadden (1982) pointed out, the significance levels in the calculations by McElhinny's are stringent. It is quite likely that the A magnetization was acquired prior to the tilting of the beds. A similar argument for the Spout Falls B leads to the conclusion that this magnetization was probably acquired post-tilting. Knight (1983) suggested that the age of the main deformation of the sub-basin was Late Pennsylvanian or younger and hence the Spout Falls B is probably representative of Kiaman magnetization. The major arguments in favor of recognition of Spout Falls B as Kiaman are the reversed nature of this magnetization, and the better grouping of the directions before applying correction for tilt. Further, the Spout Falls B direction is identical with the earlier recognized Kiaman magnetizations from Newfoundland, viz. Deer Lake H (Irving and Strong, 1984b) and Terreneville H (Kent, 1982) magnetizations. For the magnetization A of the Jeffreys Village Member (Table III), the slight improvement in grouping with tilt correction suggests that the magnetization was acquired before tilting of the beds. The bottom bed of the Jeffreys Village Member collection is reversely magnetized with respect to the rest of the beds. This reversal is present in all the specimens from that bed and is probably representative of a geomagnetic reversal thus reinforcing the primary

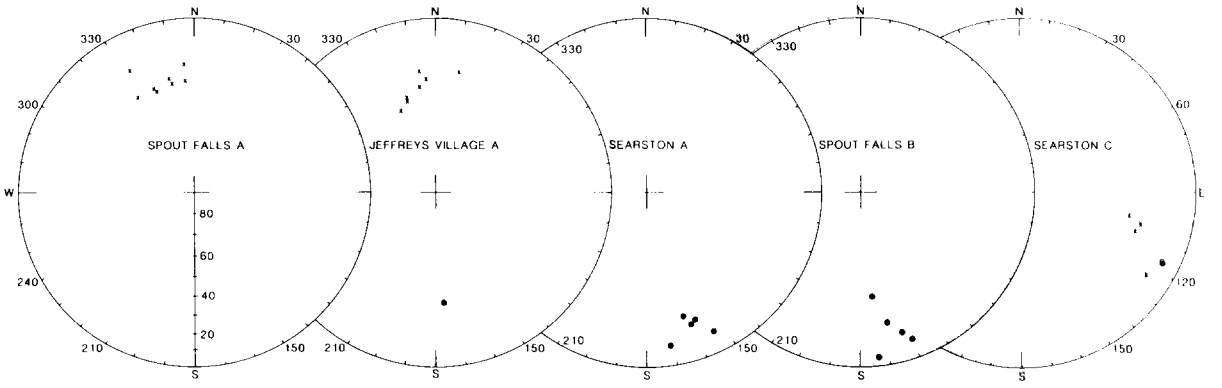


Fig. 8. Site mean directions for all isolated magnetizations. Solid symbols (crosses) indicate positive (negative) inclinations.

nature of this magnetization, of Middle–Upper Visean age. The Searston Formation exhibits two magnetizations labelled A and C (Table IV). On the same criteria as before, it may be concluded that magnetization A was probably acquired before tilting of the beds with significant improvement in the precision parameter. Magnetization C is probably post-tilting although the change in k for this magnetization is very marginal with tilt correction. The site mean directions for all these magnetizations are plotted in Fig. 8.

The magnetizations obtained in this study may be compared with the earlier preliminary studies of some of the formations by Nairn et al. (1959), Black (1964) and Opdyke (1981). Nairn et al.'s collection is probably of the Jeffreys Village Member as judged by their description of locality and geology. They reported a reverse magnetization for

TABLE IV

Isolated magnetizations for the Searston Formation

Site	N	n	n'/n	Insitu				Tilt corrected			
				D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)
Magnetization A											
Q	5	10	10/10	178.2	64.4	50.3	10.9	160.6	13.1	293	4.5
R	3	6	6/6	180.5	72.4	199	17.8	158.8	14.0	1910	5.7
S	4	8	8/8	196.2	58.4	69.5	11.1	171.5	7.0	91.1	9.7
T	4	8	8/8	159.3	65.6	25.2	18.6	153.9	7.0	91.1	9.7
U	3	6	6/6	165.9	33.4	97.4	12.6	163.6	17.1	34.0	21.5
Magnetization B											
Mean	5 sites			175.0	59.6	23.7	16.0	161.7	11.7	107	7.4
Q	5	10	10/10	105.3	−19.7	8.0	28.8	69.4	−50.6	9.7	25.9
R	3	6	6/6	108.8	−20.5	5.0	36.2	63.5	−51.4	4.2	39.4
S	4	8	8/8	116.2	5.8	7.9	35.0	102.2	−40.4	8.0	34.8
T	4	8	8/8	124.0	−8.4	17.4	22.7	93.0	−57.3	10.0	30.8
U	3	6	6/6	102.3	−25.5	7.3	49.5	95.3	−36.6	8.8	44.2
Mean	5 sites			111.6	−13.8	28.8	14.5	85.8	−48.3	32.36	13.7

N = number of samples; n = number of specimens used in demagnetization; n'/n = number of specimens showing the magnetization/number of specimens used in demagnetization; D = declination; I = inclination; k = Fisher's precision parameter; α_{95} = radius of the 95% circle of confidence.

TABLE V

Summary of Paleomagnetic results from the Carboniferous of Newfoundland, east coast basin and interior North America

Magnetization	Probable age	D (°)	I (°)	k	α_{95} (°)	P lat., P long. (°)	dp, dm (°)	Paleolat. (°)	References
Newfoundland									
Searston C	??	111.6	−13.8	28.8	14.5	19.6 S, 19.0 E	7.6, 14.8	7.0 N ± 8	This study
Spout Falls B	Late Pennsylvanian	166.7	12.2	51.7	10.7	34.5 S, 42.7 W	5.5, 10.9	6.2 N ± 6	This study
Deer Lake H	Late Pennsylvanian	157.8	−13.3	37	10.1	45.4 N, 154.6 E	5.3, 10.3	7 N ± 5	Irving and Strong (1984a)
Terrenceville H	Late Pennsylvanian	185.9	3.3	61	7.2	43.6 N, 117.1 E	3.6, 7.2	2 S ± 4	Kent (1982)
Spanish Room Fm.	Permo–Carboniferous	177	−1		5				
St. Lawrence pluton	??	177	−1		11	47 N, 129 E	5, 9	3 N ± 5	Irving and Strong (1984c)
Searston A	Namurian	161.7	11.7	107	7.4	33.9 S, 37.2 W	3.8, 7.5	5.9 S ± 4	This study
Jeffreys Vill. A	Middle–Upper Visean	351.2	−27.3	54.0	7.6	26.5 N, 130.7 E	4.5, 8.3	14.8 S ± 4	This study
Deer Lake Group	Visean	354.5	−35.5	27	10.8	21.2 N, 128.0 E	7.2, 12.5	20 S ± 7	Irving and Strong (1984a)
Terrenceville	Tournaisian	181.6	28.0	27	10.1	27.4 N, 123.6 E	6.1, 11.1	15 S ± 6	Kent (1982)
Spout Falls A	Tournaisian	343.5	−22.7	61.2	7.1	28.6 N, 139.5 E	4.5, 8.5	11.6 S ± 4	This study
East coast basin									
P.E.I. red beds	Permo–Carboniferous	168	0	36	6	42 N, 133 E	3, 6	0 ± 3	Roy (1966)
Tormentine Fm.	Upper Pennsylvanian	168	4	107	4	41 N, 132 E	2, 4	2 S ± 2	Roy (1966)
Morien Group	Late Carboniferous	171	6	32	6	40 N, 131 E	3, 6	3 S ± 3	Scotese et al. (1984)
Bonaventure Fm.	Carboniferous	166	5	20	10	38 N, 133 E	5, 10	2 S ± 5	Roy (1966)
Hurley Creek Fm.	Pennsylvanian	171	9	58	10	39 N, 125 E	5, 10	4 S ± 5	Roy et al. (1968)
Cumberland Group	Pennsylvanian	172	16	85	5	36 N, 125 E	3, 5	8 S ± 3	Roy (1969)
Riversdale Group	Lower Pennsylvanian	175	15	91	6	36 N, 122 E	3, 6	8 S ± 3	Roy (1977)
Maringouin and Shepody Fm.	Mississippi–Pennsylvanian boundary	174	15	89	3	36 N, 122 E	2, 3	8 S ± 2	Roy and Park (1974)
Windsor Group	Visean	163	14	118	6	36 N, 137 E	3, 6	7 S ± 3	Scotese et al. (1984)
Horton Group	Tournaisian	163	22	87	8	32 N, 136 E	5, 9	11 S ± 5	Scotese et al. (1984)
Interior North America [1]									
Casper Formation	Late Pennsylvanian	145	−19	40	2	46 N, 129 E	1, 2	10 N ± 1	Diehl and Shive (1981)
Bonna Terre Fm.	Late Pennsylvanian	154	7	120	5	43 N, 126 E	3, 5	4 S ± 3	Wisniowiecki et al. (1983)
Red Mountain Fm.	Late Carboniferous	150	20	64	4	38 N, 132 E	2, 4	10 S ± 2	Perroud and Vander Voo (1984) Hodych et al. (1984)
Churchland pluton	282 + 6 Ma	337	−30	4	16	34 N, 126 E	10, 18	16 S ± 10	Barton and Brown (1983)
Buffalo siltstone [2]	Upper Pennsylvanian					27 S, 57 W	α_{95} = 6.1		Payne et al. (1981)
Brush Creek 1st. [2]	Upper Pennsylvanian					36 S, 56 W	α_{95} = 4.2		Payne et al. (1981)
Mauch Chunk Fm.	Lower Carboniferous	data not available						15 S	Kent and Opdyke (1984)
New Brunswick volc.	Tournaisian/Namurian	161	11	78	6	36 N, 136 E	3, 6	6 S ± 3	Seguin (1984)
St. Joe 1st.	Early Tournaisian	149	8	218	2	39 N, 132 E	1, 2	4 S ± 1	Scott (1979)
Barnett Formation	Mississippian	156	6	905	3	49 N, 119 E	2, 3	3 S ± 2	Kent and Opdyke (1979)
Lilesville pluton	326 ± 27 Ma	334	−16	41	3	40 N, 134 E	2, 3	8 S ± 2	Barton and Brown (1983)
Austell gneiss		155	30			34 N, 126 E	7, 13	16 S ± 7	Ellwood and Abrams (1982)

P lat., P long. = latitude and longitude of pole; dp, dm = semi-axes of the oval of 95% confidence; Paleolat. = paleolatitude of the Formation; rest of symbols have same meaning as in Table II.

[1] Irving and Strong (1984b) showed that for many of the cratonic data, the Kiaman magnetization has not been resolved from the primary;

[2] data on directions are not available in the original.

nine samples (D = 166, $I = 8$) and only at the NRM level. Black's (1964) collection includes 32 samples from "the Mississippian Codroy Group from the southeast shore of St. Georges Bay", demagnetized to 30 mT alternating field. Mean direction reported was at $D = 174.5°$, $I = 23.4°$. No mention was made of any structural corrections. It is possible, because of inadequate demagnetization in their experimentation, the results reported by Nairn et al. and Black are only Kiaman overprints. Opdyke (1981) reported (in abstract form) thermally demagnetized results from 50 samples of the "Barachois Fm." yielding a "characteristic magnetization directed north–northeast and upward with a possible reversal also present". It is not clear what the "Barachois Fm." of Opdyke correlates with in the collections of the present study. If it is the Searston Formation, the present results differ from Opdyke's in that the Searston Formation exhibited only reversed magnetization while Opdyke's abstract refers to "possible presence of reversal". However, because of insufficient information, it is uncertain whether results from the same formation are compared. Table V lists, along with other data, the mean magnetizations obtained in the present study, their probable ages of acquisition, the corresponding paleopoles and paleolatitudes of the site.

5. Discussion

The paleopoles and the corresponding paleolatitudes for the Bay St. George sub-basin may be compared with those units of the corresponding age from the east coast basin and also the Carboniferous of interior North America (Table V). It is to be borne in mind, however, in these comparisons, that although the Bay St. George sub-basin is in western Newfoundland, there is no geological evidence that it rests directly on the basement as in the case of the Deer Lake basin to the north. Thus, although it is likely that the data presented here refer to the terrane on the cratonic side of the Appalachians, there is no unambiguous geological evidence for such statement. Firstly, the paleolatitude of the Spout Falls A agrees well with that of the Tournaisian Terrenceville Formation of eastern Newfoundland studied earlier by Kent (1982). The paleolatitude of Jeffreys Village A agrees well with that of the Visean Deer Lake Group of western Newfoundland (Irving and Strong, 1984b) which rests directly on the Grenville basement. Such agreements confirm the earlier conclusion by Irving and Strong (1984b) that movements by large distances (hundreds of kilometres) between east and west Newfoundland since the Early Carboniferous are unlikely. The paleolatitudes of the Bay St. George sub-basin Formations agree well with those of the corresponding units of the east coast basin formations. This suggests that there is no north–south relative movement between the Maritime basin and the Bay St. George sub-basin. It may be seen in Table V that there is a paleolatitudinal change from about 15° in Early Carboniferous to near equatorial paleolatitudes by Late Pennsylvanian times. This is probably the result of a northerly displacement of the Appalachians *and* North America during the Carboniferous.

Roy and Morris (1983), in their elaborate discussion of the Carboniferous paleomagnetism, suggested that a major change in the global geomagnetic field behavior took place during the Carboniferous from a period of both normal and reversed magnetizations to a period of constantly reversed polarity interval (the Kiaman interval). They proposed a magnetostratigraphic horizon marker for this change at the Namurian–Westphalian boundary and placed this marker for the east coast basin at the top of the Riversdale Group level and base of the Cumberland Group. It is of interest to see if this placement of the horizon marker agrees with the results from the Bay St. George sub-basin. The Bay St. George sub-basin results are consistent with the general concept of a normal (and possibly some reversed) magnetization followed by only a reversed magnetization during Late Carboniferous. The older Spout Falls A and Jeffreys Village A magnetizations are dominantly normal followed by reversed magnetization of the Searston A. The horizon marker in the Bay St. George sub-basin can probably be placed at the base of the Searston Formation (Barachois Group). On the basis of magnetostratigraphy, the Barachois Group may be correlatable with the

Cumberland Group of the east coast basin. It should be possible to place the horizon marker more precisely by a more thorough, continuous and systematic sampling of the Codroy and Barachois Groups and particularly at the boundaries.

Scotese et al. (1984) recently obtained paleopoles for the Horton and Windsor Groups of Nova Scotia which are considered from stratigraphic correlations to be equivalent to Anguille and Codroy Groups, respectively. It is interesting that the data for Horton and Windsor Groups is 100% reversed while for the Spout Falls Formation and the Jeffreys Village Member, they are dominantly normal. It is likely that all these Groups have both normal and reverted magnetizations and sampling, by chance, in the Windsor and Horton Groups, was limited to one particular polarity. Scotese et al.'s (1984) results are included in Table V.

Kent and Opdyke (1978, 1979, 1980) argued, on the basis of comparison of Devonian and Carboniferous results from cratonic North America and from northern Appalachians, that the Maritime–New England region of Canada and U.S. is a displaced terrane which was situated ~ 1500 km south (relative to cratonic North America) in the Early Carboniferous and which got welded to North America after a sinistral motion which ceased in Upper Carboniferous/Permian. As mentioned earlier, Roy (1982) and more recently Irving and Strong (1984c) criticized the basis for this hypothesis on the grounds that cratonic paleomagnetic data used were not representative of the claimed ages and an unresolved Kiaman magnetization was present in the cratonic Devonian and Early Carboniferous data. Irving and Strong (1984b) further showed that their Deer Lake Group data of western Newfoundland is representative of Visean magnetization for cratonic rocks. This agrees well with the "Acadian" Terrenceville data (Tournaisian) and discounts any hypothesis involving displaced terranes. The results from the Carboniferous Bay St. George basin have a bearing on the displaced terrane hypothesis. The results suggest that there were no relative movements since the Early Carboniferous between the east coast basin and Bay St. George sub-basin and also between eastern and western Newfoundland. If the ~ 1500 km sinistral motion, as originally proposed by Kent and Opdyke, took place before the Carboniferous these results would have no bearing on the problem. If the hypothesis, as discussed by various authors, includes part of the motion taking place since the beginning of the Carboniferous (as suggested for example by Diehl and Shive, 1981), the Bay St. George results do not show any evidence supporting such motion. If the displaced terrane hypothesis is still a tenable one, then, western Newfoundland must be part of "Acadia" of Kent and Opdyke and the fault about which the sinistral motion took place must be west of western Newfoundland. The presence of such a fault or evidence for it is unknown at the moment. It is of interest to note that Kent and Opdyke (1984) recently re-studied the Mauch Chunk Formation in the U.S. Central Appalachians and, based on the results, partly discounted the hypothesis of "Acadia".

6. Conclusions

(1) The magnetostratigraphic horizon marker separating the older dominant normal magnetization from the younger reversed magnetization during the Carboniferous may be placed at the base of the Searston Formation for the Bay St. George sub-basin. From plant spores data, this is near the Visean/Namurian boundary.

(2) The data suggest that no large-scale relative movement took place between eastern and western Newfoundland since the Early Carboniferous.

(3) No north–south relative movement is indicated between the east coast basin and the Bay St. George sub-basin since the Carboniferous.

(4) The paleolatitude of western Newfoundland changes from about 15° to near equatorial value from Early to Late Carboniferous. This change, which can also be seen in the Maritime and in the Interior North American data, suggests a northward motion of the Appalachians *and* North America as a whole during the Carboniferous.

(5) Predominantly normal characteristic magnetizations obtained for the Spout Falls and the Robinsons River Formations (Jeffreys Village

Member) differ only in polarity from the entirely reversed magnetization obtained for the Horton and Windsor Groups by other workers.

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